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Database Project : ArtConnect Application

Step 2

1) Identify entities and relationships

We added ID field to every entity for easier use

-Artists :ID, name, bio, birthyear, list disciplines, contactEmail, phone, city, website, socialMedia, IsActive, list Artworks

Functional dependencies : every attribute depends on the ID of the artists.

-Artwork : id, title, creationYear, type, medium, dimensions, description, price, status, artist, tags

Functional dependencies : id $\rightarrow$ title, creationYear, type, medium, dimensions, description, price, status, artist, tags

- ArtworkTag : name

Functional dependencies : no dependencies (only one field)

- Booking : id, workshop, member, bookingDate, paymentStatus

Functional dependencies : id $\rightarrow$ workshop, member, bookingDate, paymentStatus

- Community member : ID, name, email, birthyear, phone, city, list favoriteDisciplines, membershipType, bookings, reviews

Functional Dependencies : all the attributes depends on the ID

- Discipline : name

Functional dependencies : no dependencies (only one field)

- Exhibition : ID, Title, startDate, endDate, description, gallery, curatorName, theme, list Artworks

Functional Dependencies : no dependencies except on the ID

- Gallery : ID, name, address, ownerName, openingHour, contactPhone, rating, website, list exhibition

Functional Dependencies : no dependencies except on the ID

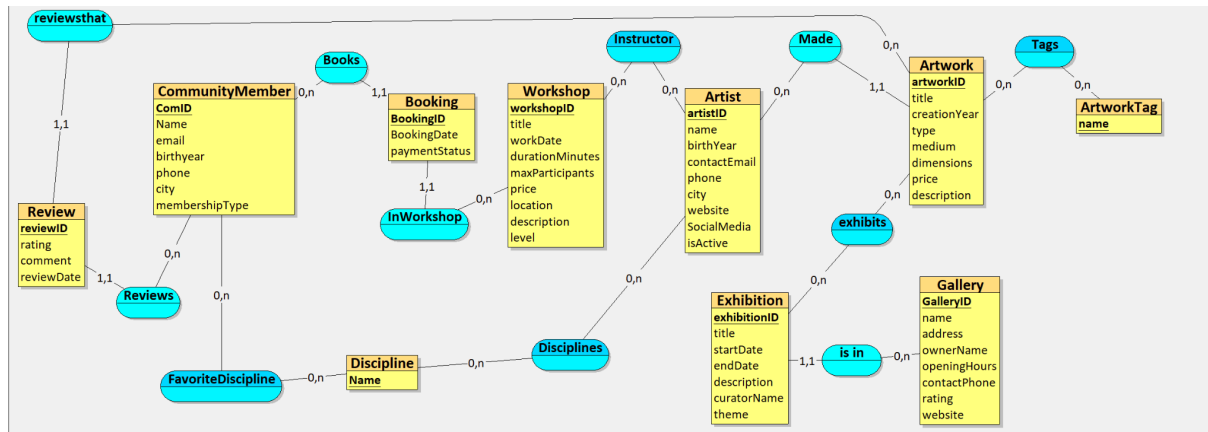
- Review : ID, reviewer, artwork, rating, comment, reviewDate

Functional Dependencies : no dependencies except on the ID

- Workshop : id, reviewer, artwork, rating, comment, reviewDate

Functional Dependencies : no dependencies except on the ID

2.



3.

CommunityMember = (**ComID INT**, Name VARCHAR(50), email VARCHAR(50), birthyear INT, phone VARCHAR(50), city VARCHAR(50), membershipType VARCHAR(50));

Discipline = (**discName VARCHAR(50)**);

Workshop = (**workshopID INT**, title VARCHAR(50), workDate DATE, durationMinutes INT, maxParticipants INT, price DOUBLE, location VARCHAR(50), description VARCHAR(50), level VARCHAR(50));

Artist = (**artistID INT**, name VARCHAR(50), birthYear INT, contactEmail VARCHAR(50), phone VARCHAR(50), city VARCHAR(50), website VARCHAR(50), SocialMedia VARCHAR(50), isActive LOGICAL);

Artwork = (**artworkID INT**, title VARCHAR(50), creationYear INT, type VARCHAR(50), medium VARCHAR(50), dimensions VARCHAR(50), price DOUBLE, description VARCHAR(50), #artistID);

Gallery = (**GalleryID INT**, name VARCHAR(50), address VARCHAR(50), ownerName VARCHAR(50), openingHours VARCHAR(50), contactPhone VARCHAR(50), rating DOUBLE, website VARCHAR(50));

ArtworkTag = (**name VARCHAR(50)**);

Booking = (**BookingID INT**, BookingDate DATE, paymentStatus VARCHAR(50), #workshopID, #ComID);

Review = (**reviewID INT**, rating INT, comment VARCHAR(50), reviewDate DATE, #artworkID, #ComID);

Exhibition = (**exhibitionID INT**, title VARCHAR(50), startDate DATE, endDate DATE, description VARCHAR(50), curatorName VARCHAR(50), theme VARCHAR(50), #GalleryID);

FavoriteDiscipline = (**#ComID, #discName**);

Disciplines = (#discName, #artistID);

Instructor = (#workshopID, #artistID);

exhibits = (#artworkID, #exhibitionID);

Tags = (#artworkID, #name);

```
4.CREATE TABLE CommunityMember(  
    ComID INT,  
    Name VARCHAR(50),  
    email VARCHAR(50),  
    birthyear INT,  
    phone VARCHAR(50),  
    city VARCHAR(50),  
    membershipType VARCHAR(50),  
    PRIMARY KEY(ComID)  
);
```

```
CREATE TABLE Discipline(  
    discName VARCHAR(50),  
    PRIMARY KEY(discName)  
);
```

```
CREATE TABLE Workshop(  
    workshopID INT,  
    title VARCHAR(50),  
    workDate DATE,  
    durationMinutes INT,  
    maxParticipants INT,  
    price DOUBLE,  
    location VARCHAR(50),  
    description VARCHAR(50),  
    level VARCHAR(50),  
    PRIMARY KEY(workshopID)  
);
```

```
CREATE TABLE Artist(  
    artistID INT,  
    name VARCHAR(50),  
    birthYear INT,  
    contactEmail VARCHAR(50),  
    phone VARCHAR(50),  
    city VARCHAR(50),  
    website VARCHAR(50),  
    SocialMedia VARCHAR(50),
```

```
    isActive LOGICAL,  
    PRIMARY KEY(artistID)  
);
```

```
CREATE TABLE Artwork(  
    artworkID INT,  
    title VARCHAR(50),  
    creationYear INT,  
    type VARCHAR(50),  
    medium VARCHAR(50),  
    dimensions VARCHAR(50),  
    price DOUBLE,  
    description VARCHAR(50),  
    artistID INT NOT NULL,  
    PRIMARY KEY(artworkID),  
    FOREIGN KEY(artistID) REFERENCES Artist(artistID)  
);
```

```
CREATE TABLE Gallery(  
    GalleryID INT,  
    name VARCHAR(50),  
    address VARCHAR(50),  
    ownerName VARCHAR(50),  
    openingHours VARCHAR(50),  
    contactPhone VARCHAR(50),  
    rating DOUBLE,  
    website VARCHAR(50),  
    PRIMARY KEY(GalleryID)  
);
```

```
CREATE TABLE ArtworkTag(  
    name VARCHAR(50),  
    PRIMARY KEY(name)  
);
```

```
CREATE TABLE Booking(  
    BookingID INT,  
    BookingDate DATE,  
    paymentStatus VARCHAR(50),  
    workshopID INT NOT NULL,  
    ComID INT NOT NULL,  
    PRIMARY KEY(BookingID),  
    FOREIGN KEY(workshopID) REFERENCES Workshop(workshopID),  
    FOREIGN KEY(ComID) REFERENCES CommunityMember(ComID)  
);
```

```
CREATE TABLE Review(  
    reviewID INT,
```

```
rating INT,  
comment VARCHAR(50),  
reviewDate DATE,  
artworkID INT NOT NULL,  
ComID INT NOT NULL,  
PRIMARY KEY(reviewID),  
FOREIGN KEY(artworkID) REFERENCES Artwork(artworkID),  
FOREIGN KEY(ComID) REFERENCES CommunityMember(ComID)  
);
```

```
CREATE TABLE Exhibition(  
exhibitionID INT,  
title VARCHAR(50),  
startDate DATE,  
endDate DATE,  
description VARCHAR(50),  
curatorName VARCHAR(50),  
theme VARCHAR(50),  
GalleryID INT NOT NULL,  
PRIMARY KEY(exhibitionID),  
FOREIGN KEY(GalleryID) REFERENCES Gallery(GalleryID)  
);
```

```
CREATE TABLE FavoriteDiscipline(  
ComID INT,  
discName VARCHAR(50),  
PRIMARY KEY(ComID, discName),  
FOREIGN KEY(ComID) REFERENCES CommunityMember(ComID),  
FOREIGN KEY(discName) REFERENCES Discipline(discName)  
);
```

```
CREATE TABLE Disciplines(  
discName VARCHAR(50),  
artistID INT,  
PRIMARY KEY(discName, artistID),  
FOREIGN KEY(discName) REFERENCES Discipline(discName),  
FOREIGN KEY(artistID) REFERENCES Artist(artistID)  
);
```

```
CREATE TABLE Instructor(  
workshopID INT,  
artistID INT,  
PRIMARY KEY(workshopID, artistID),  
FOREIGN KEY(workshopID) REFERENCES Workshop(workshopID),  
FOREIGN KEY(artistID) REFERENCES Artist(artistID)  
);
```

```
CREATE TABLE exhibits(  

```

```
artworkID INT,  
exhibitionID INT,  
PRIMARY KEY(artworkID, exhibitionID),  
FOREIGN KEY(artworkID) REFERENCES Artwork(artworkID),  
FOREIGN KEY(exhibitionID) REFERENCES Exhibition(exhibitionID)  
);
```

```
CREATE TABLE Tags(  
artworkID INT,  
name VARCHAR(50),  
PRIMARY KEY(artworkID, name),  
FOREIGN KEY(artworkID) REFERENCES Artwork(artworkID),  
FOREIGN KEY(name) REFERENCES ArtworkTag(name)  
);
```